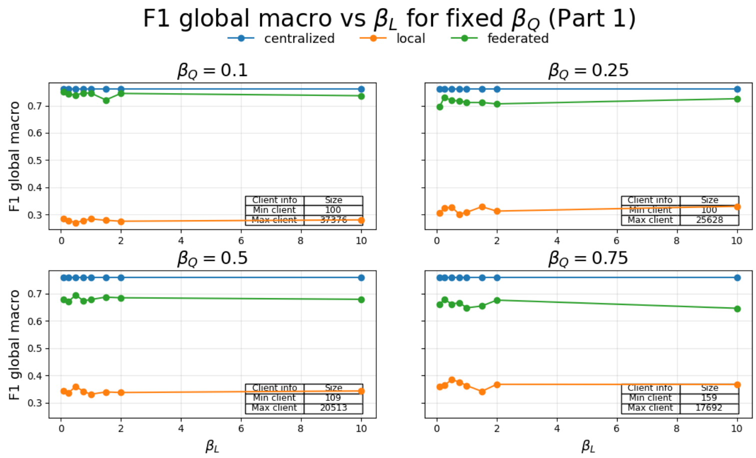
Beta Sweep Results

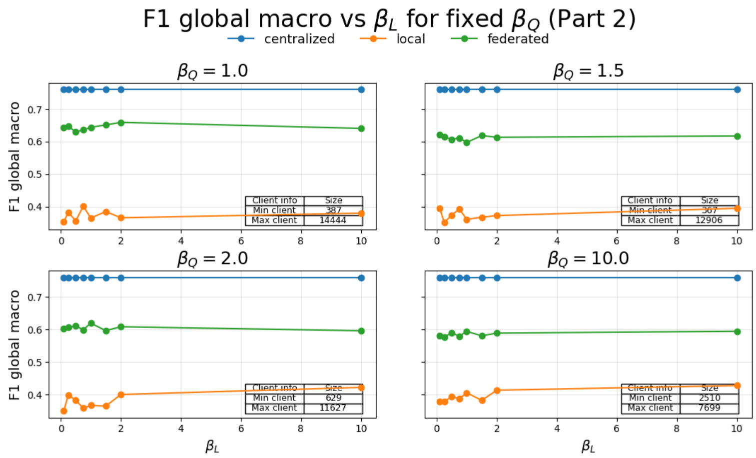
Label Skew

We have run a grid search/beta sweep. Using seed 42, when the values:

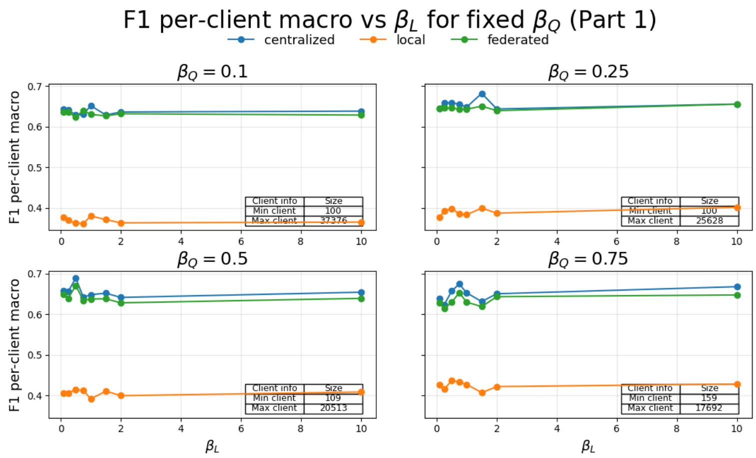
$$\beta_Q = [0.1, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 10.0]$$
$$\beta_L = [0.1, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 10.0]$$

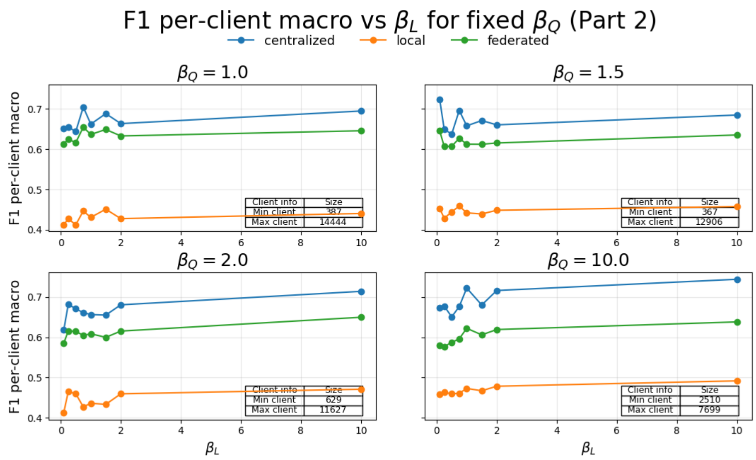
The eight plots below show the F1 macro metric (computed using global thresholds) as a function of label skew (β_L) for a fixed quantity skew (β_Q).



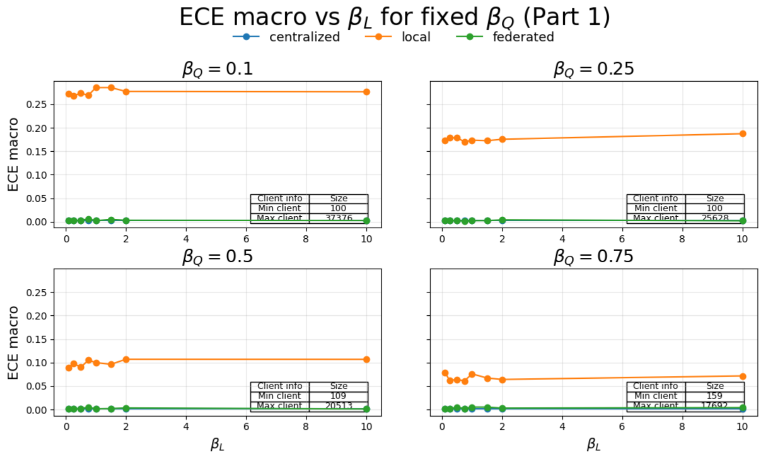


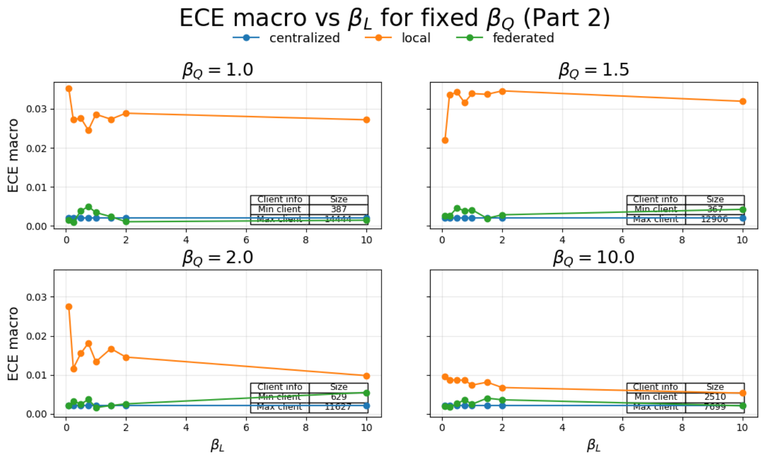
The results show a clear performance ranking between the three training paradigms. Centralized training consistently achieves the highest F1 score, followed by federated learning, while locally trained models perform substantially worse. This confirms that federated learning successfully leverages information across clients, significantly outperforming purely local training.
Varying the label skew parameter β_L has only a minor impact on model performance. The curves remain relatively flat across all levels of label skew, suggesting that the model and training procedure are robust to moderate levels of label distribution heterogeneity.
High quantity skew results in a high preformance of the federated model. This is likely due to that few/one client has most of the datapoints resulting in a model close to the centralized model.
Below is similar plots with F1 macro score but with per client threads holds:



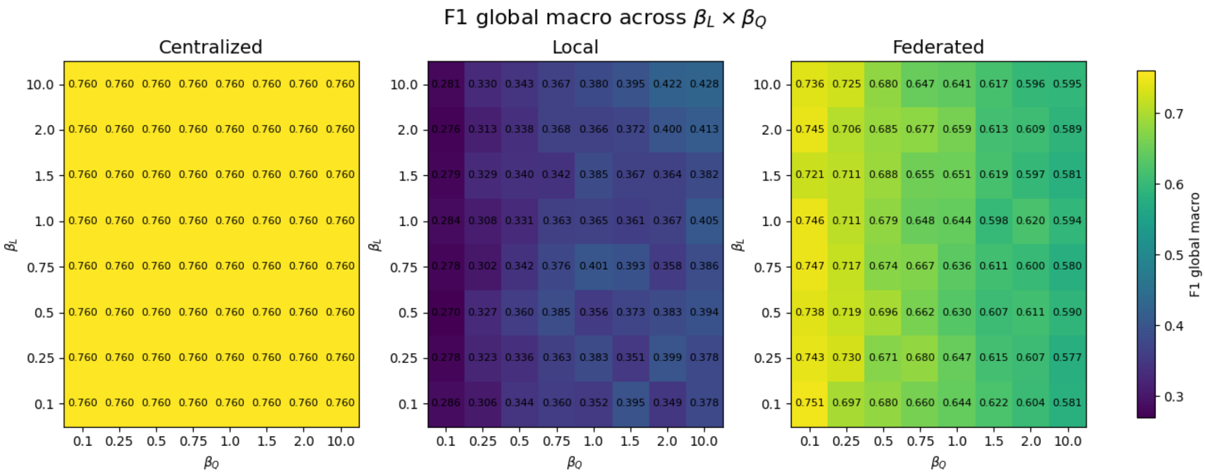


The per-client F1 scores are generally lower than the corresponding global F1 scores. Despite this difference in scale, the overall trends remain consistent across both metrics. Centralized training achieves the highest performance, followed by federated learning, while locally trained models perform substantially worse. Furthermore, the influence of label skew remains limited, whereas quantity skew has a somewhat larger effect on federated performance.
Below is simelar plots with ece metric:





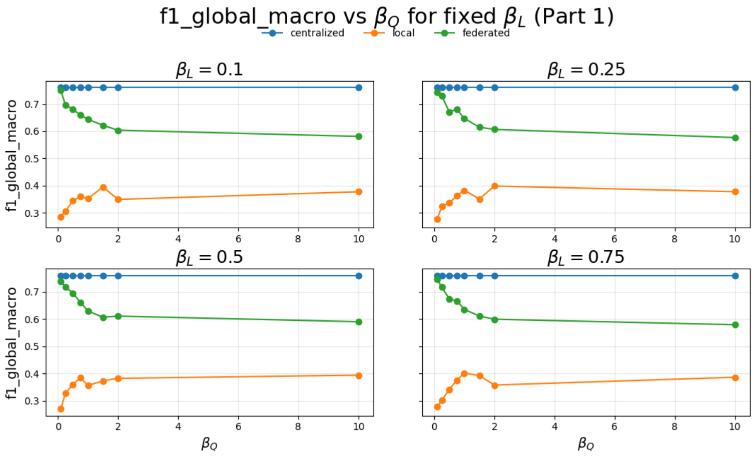
Again we see the same pattern. Finally we have made an interaktion plot:

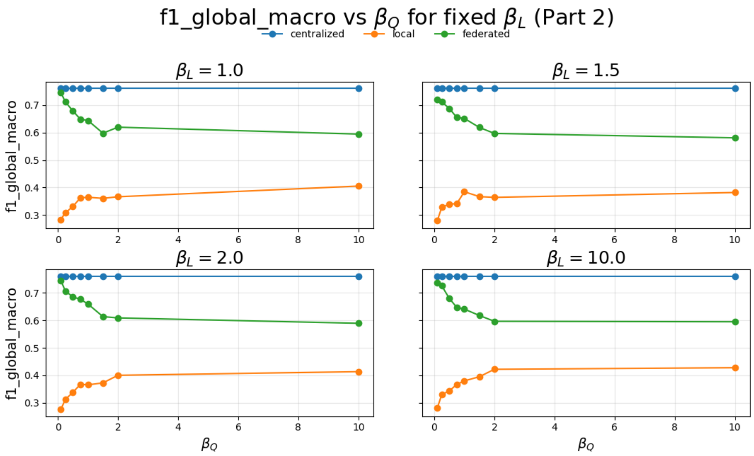


Again we see that the federated model perform worst with high quantity skew, and is relativly robust to label skew.

0.0.1 Quantity Skew

The eight plots below show the F1 macro metric (computed using global thresholds) as a function of quantity skew (β_Q) for a fixed label skew (β_L).

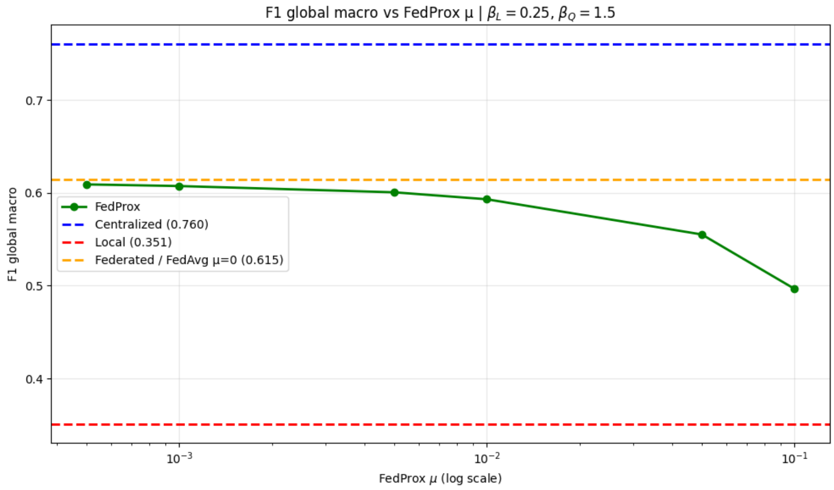


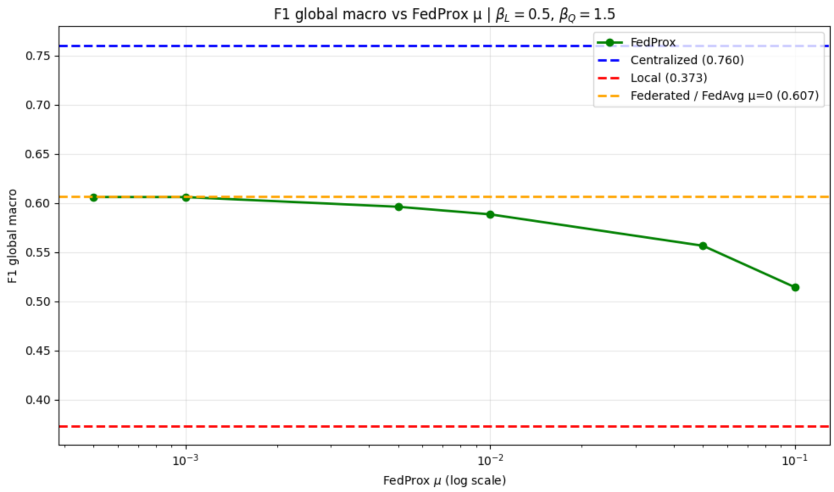


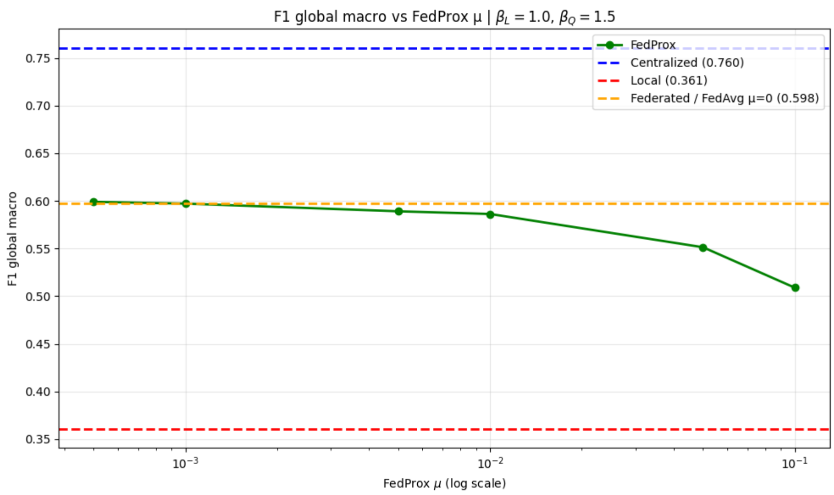
The plots show the same pattern, that client size has a big impact on model performance. Since data where one/few clients has most of the data points are close to the centralized model.

0.0.2 FedProx

Below is three plots of the FedProx model. The x-axis has the μ-value. For reference we have centralized, local and FedAvg show by dashed lines. In the three plots we have used a moderate quantity skew of $\beta_Q = 1.5$ where minimum client size is 307 and maximum client size is 12906. In the three plot we vary the label skew with the values 0.25,0.5 and 1.0:







These findings indicate that the additional proximal constraint introduced by FedProx may be unnecessary in scenarios where client heterogeneity is relatively moderate.